

Threat Hunting Analysis — Compromised Machine Identification

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This notebook uses Python and Pandas to analyze forensic artifact data (parquet files) collected from two Windows hosts, in order to identify a compromised machine and supporting Indicators of Compromise (IOCs).

```
In [1]: import pandas as pd

dataset_path = r'C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc'
df = pd.read_parquet(dataset_path + '/domain_users.parquet')
print(df.head())
```

	group	username
0	seoul	timotlopez
1	seoul	timotlopez
2	seoul	timotlopez
3	domain admins	timotlopez
4	domain admins	Administrator

```
In [2]: import os

for root, dirs, files in os.walk(dataset_path):
    for f in files:
        print(os.path.join(root, f))
```

```
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\DS_Store
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\domain_users.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\loggedonusers\0000DQEE.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\loggedonusers\0001LXQEN.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\useraccounts\0000DQEE.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\useraccounts\0001LXQEN.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32drivers\0000DQEE.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32drivers\0001LXQEN.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32persistence-fileitems\0000DQEE.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32persistence-fileitems\0001LXQEN.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32persistence-registryitems\0000DQEE.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32persistence-registryitems\0001LXQEN.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32persistence-servicesitems\0000DQEE.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32persistence-servicesitems\0001LXQEN.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32processes\0000DQEE.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32processes\0001LXQEN.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32processes-memorysections\0000DQEE.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32processes-memorysections\0001LXQEN.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32services\0000DQEE.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32services\0001LXQEN.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32tasks\0000DQEE.parquet
C:\Users\nahla\Desktop\threat_hunt_task\3d276079-8448-42c6-a394-bd54c981eedc\w32tasks\0001LXQEN.parquet
```

```
In [3]: import glob

def load_all(folder_name):
    files = glob.glob(dataset_path + f'/{folder_name}/*.parquet')
    dfs = []
    for f in files:
        d = pd.read_parquet(f)
        d['hostname'] = os.path.basename(f).replace('.parquet', '')
        dfs.append(d)
    return pd.concat(dfs, ignore_index=True) if dfs else pd.DataFrame()
```

```
In [4]: processes = load_all('w32processes')
processes_memory = load_all('w32processes-memorysections')
services = load_all('w32services')
tasks = load_all('w32tasks')
drivers = load_all('w32drivers')
persistence_reg = load_all('w32persistence-registryitems')
persistence_files = load_all('w32persistence-fileitems')
persistence_svc = load_all('w32persistence-serviceitems')
logged_users = load_all('loggedonusers')
user_accounts = load_all('useraccounts')
```

```
print(processes.shape, processes_memory.shape, services.shape, tasks.shape)
```

```
(95, 6) (3274, 9) (997, 14) (253, 14)
```

```
In [5]: print(logged_users.columns.tolist())
logged_users
```

```
['domain-user', 'hostname', 'local-user', 'username']
```

```
Out[5]:
```

	domain-user	hostname	local-user	username
0	False	0000DQQUEE	True	svchost.exe
1	False	0000DQQUEE	True	winlogon.exe
2	False	0000DQQUEE	True	RuntimeBroker.exe
3	False	0000DQQUEE	True	lsass.exe
4	False	0000DQQUEE	True	dwm.exe
5	False	0000DQQUEE	True	conhost.exe
6	False	0000DQQUEE	True	LogonUI.exe
7	False	0000DQQUEE	True	spoolsv.exe
8	False	0000DQQUEE	True	dllhost.exe
9	False	0000DQQUEE	True	msdtc.exe
10	False	0000DQQUEE	True	taskhostw.exe
11	False	0000DQQUEE	True	sihost.exe
12	False	0000DQQUEE	True	Explorer.EXE
13	False	0000DQQUEE	True	TabTip32.exe
14	False	0000DQQUEE	True	MpCmdRun.exe
15	False	0000DQQUEE	True	ShellExperienceHost.exe
16	False	0000DQQUEE	True	SearchUI.exe
17	True	0000DQQUEE	False	timotlopez
18	False	0000DQQUEE	True	AvastUI.exe
19	False	0000DQQUEE	True	AvastSvc.exe
20	False	0000DQQUEE	True	OneDrive.exe
21	False	0000DQQUEE	True	ieexplore.exe
22	True	0000DQQUEE	False	svc_lw
23	False	0000DQQUEE	True	cmd.exe
24	False	0000DQQUEE	True	kitty-0.70.0.6.exe
25	False	0000DQQUEE	True	CyberGhost.Service.exe
26	False	0000DQQUEE	True	putty.exe
27	False	0000DQQUEE	True	Calculator.exe
28	False	0000DQQUEE	True	net.exe
29	False	0000DQQUEE	True	reg.exe
30	False	0001LXQEN	True	svchost.exe
31	False	0001LXQEN	True	winlogon.exe
32	False	0001LXQEN	True	RuntimeBroker.exe
33	False	0001LXQEN	True	lsass.exe
34	False	0001LXQEN	True	dwm.exe
35	False	0001LXQEN	True	conhost.exe
36	False	0001LXQEN	True	LogonUI.exe
37	False	0001LXQEN	True	spoolsv.exe
38	False	0001LXQEN	True	dllhost.exe
39	False	0001LXQEN	True	msdtc.exe
40	False	0001LXQEN	True	taskhostw.exe
41	False	0001LXQEN	True	sihost.exe

42	False	0001LXQEN	True	Explorer.EXE
43	False	0001LXQEN	True	TabTip32.exe
44	False	0001LXQEN	True	MpCmdRun.exe
45	False	0001LXQEN	True	ShellExperienceHost.exe
46	False	0001LXQEN	True	SearchUI.exe
47	True	0001LXQEN	False	timotlopez
48	False	0001LXQEN	True	AvastUI.exe
49	False	0001LXQEN	True	AvastSvc.exe
50	False	0001LXQEN	True	OneDrive.exe
51	False	0001LXQEN	True	iexplore.exe
52	False	0001LXQEN	True	mspaint.exe
53	False	0001LXQEN	True	cmd.exe
54	False	0001LXQEN	True	ruby.exe
55	True	0001LXQEN	False	Administrator
56	False	0001LXQEN	True	CCleaner64.exe
57	False	0001LXQEN	True	CyberGhost.Service.exe
58	False	0001LXQEN	True	KeePassXC.exe

In [6]: `logged_users[logged_users.astype(str).apply(lambda row: row.str.contains('timotlopez|Administrator', case=False`

Out[6]:

	domain-user	hostname	local-user	username
17	True	0000DQEE	False	timotlopez
47	True	0001LXQEN	False	timotlopez
55	True	0001LXQEN	False	Administrator

In [7]: `print(processes.columns.tolist())`
`processes[['hostname'] + [c for c in processes.columns if 'name' in c.lower() or 'path' in c.lower()]]`
`['arguments', 'hostname', 'name', 'path', 'pid', 'username']`

Out[7]:

	hostname	hostname	name	path	username
0	0000DQEE	0000DQEE	svchost.exe	C:\Windows\System32	NT AUTHORITY\SYSTEM
1	0000DQEE	0000DQEE	AvastSvc.exe		timotlopez
2	0000DQEE	0000DQEE	dwm.exe	C:\Windows\system32	NT AUTHORITY\SYSTEM
3	0000DQEE	0000DQEE	dllhost.exe	C:\Windows\system32	NT AUTHORITY\SYSTEM
4	0000DQEE	0000DQEE	putty.exe	C:\Program Files\PuTTY	svc_lw
...	...	...	...	...	...
90	0001LXQEN	0001LXQEN	AvastSvc.exe		timotlopez
91	0001LXQEN	0001LXQEN	CyberGhost.Service.exe	C:\Program Files\CyberGhost 6	timotlopez
92	0001LXQEN	0001LXQEN	dwm.exe	C:\Windows\system32	NT AUTHORITY\SYSTEM
93	0001LXQEN	0001LXQEN	svchost.exe	C:\Windows\system32	NT AUTHORITY\SYSTEM
94	0001LXQEN	0001LXQEN	KeePassXC.exe	C:\Program Files\KeePassXC	timotlopez

95 rows × 5 columns

In [8]: `suspicious = processes[processes.astype(str).apply(lambda row: row.str.contains('Temp|AppData|ProgramData', case`
`suspicious`

Out[8]:

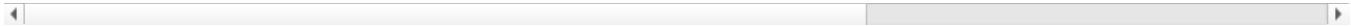
	arguments	hostname	name	path
45	"C:\Users\timotlopez\AppData\Local\Microsoft\O...	0000DQEE	OneDrive.exe	C:\Users\timotlopez\AppData\Local\Microsoft\O...
48	"C:\Users\timotlopez\AppData\Local\Microsoft\O...	0001LXQEN	OneDrive.exe	C:\Users\timotlopez\AppData\Local\Microsoft\O...
79	C:\Windows\System32\cmd.exe /C C:\Users\timotl...	0001LXQEN	cmd.exe	C:\Windows\System32\cmd.e

```
In [9]: print(persistence_reg.columns.tolist())
persistence_reg
```

['hostname', 'keypath', 'path', 'text', 'type', 'username', 'valuenamename']

Out[9]:		hostname	keypath	path	te
0	0000DQEE		Classes\CLSID\{2D3F8A1B-6DCD-4ED5-BDBA-A096594...	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{2D3...	C:\Windows\System32\twinapi...
1	0000DQEE		Classes\CLSID\{d509c21a-b88c-4ad1-8fad-d6a7572...	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{d50...	%SystemRoot%\system32\explorerframe...
2	0000DQEE		Classes\CLSID\{A6EE35C6-87EC-47DF-9F22-1D5AAD8...	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{A6E...	%SystemRoot%\system32\windowscodecs...
3	0000DQEE		Classes\CLSID\{649EEC1E-B579-4E8C-BB3B-4997F84...	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{649...	C:\Windows\System32\Dxtrans...
4	0000DQEE		Classes\CLSID\{8369AB20-56C9-11D0-94E8-00AA005...	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{836...	C:\Windows\System32\occache...
...		...	...	...	...
12168	0001LXQEN		Classes\CLSID\{B92E345D-F52D-41F3-B562-081BC77...	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{B92...	%SystemRoot%\system32\windowscodecs...
12169	0001LXQEN		Classes\CLSID\{3050f6cd-98b5-11cf-bb82-00aa00b...	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{305...	C:\Windows\System32\iepeers...
12170	0001LXQEN		Classes\CLSID\{b27b520e-46db-4720-b9c5-5f80aca...	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{b27...	%SystemRoot%\system32\hgcp...
12171	0001LXQEN		Classes\CLSID\{9456A480-E88B-43EA-9E73-0B2D9B7...	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{945...	%SystemRoot%\system32\windowscodecs...
12172	0001LXQEN		Classes\CLSID\{32DA2B15-CFED-11D1-B747-00C04FC...	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{32D...	C:\Windows\System32\scrrun...

12173 rows x 7 columns



```
In [10]: print(persistence_files.columns.tolist())
persistence_files
```

['drive', 'fileextension', 'filename', 'filepath', 'fullpath', 'hostname', 'username']

Out[10]:

	drive	fileextension	filename	filepath	
0	c	dll	pwlauncher.dll	Windows\System32	c:\W
1	c	exe	OneDrive.exe	Users\user\AppData\Local\microsoft\OneDrive	c:\Users\user\A
2	C	dll	oleaut32.dll	Windows\System32	C
3	c	sys	rdpbus.sys	Windows\System32\drivers	c:\Wind
4	c	dll	qoswmi.dll	Windows\System32\wbem	c:\Win
...	...	...	...	...	
4591	c	dll	tspubwmi.dll	Windows\System32	c:'
4592	c	dll	smartcardcredentialprovider.dll	Windows\System32	c:\Windows\System
4593	c	sys	SerCx2.sys	Windows\System32\drivers	c:\Wind
4594	c	SYS	VSTXRAID.SYS	Windows\System32\drivers	c:\Windows
4595	c	sys	rspndr.sys	Windows\System32\drivers	c:\Winc

4596 rows x 7 columns



In [11]:

```
print(persistence_svc.columns.tolist())
persistence_svc
```

[]

Out[11]: —

In [12]:

```
pd.set_option('display.max_colwidth', None)
processes[processes['pid'] == 5612]
```

Out[12]:

	arguments	hostname	name	path
79	C:\Windows\System32\cmd.exe /C C:\Users\timotlopez\AppData\Local\Microsoft\WindowsApps\run_pd.bat	0001LXQEN	cmd.exe	C:\Windows\System32\cmd.exe



In [13]:

```
persistence_reg[persistence_reg['hostname'] == '0001LXQEN']
```

Out[13]:

	hostname	keypath	path
6087	0001LXQEN	Classes\CLSID\{4be5c3b3-53a2-4f2f-af9d-d26a5327eb92}\InProcServer32	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{4be5c3b3-53a2-4f2f-af9d-d26a5327eb92}\InProcServer32\C:\Window
6088	0001LXQEN	Classes\CLSID\{ed9d80b9-d157-457b-9192-0e7280313bf0}\InProcServer32	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{ed9d80b9-d157-457b-9192-0e7280313bf0}\InProcServer32\
6089	0001LXQEN	Wow6432Node\Classes\CLSID\{3F037241-414E-11D1-A7CE-00A0C913F73C}\InProcServer32	HKEY_LOCAL_MACHINE\Software\Wow6432Node\Classes\CLSID\{3F037241-414E-11D1-A7CE-00A0C913F73C}\InProcServer32\
6090	0001LXQEN	Classes\CLSID\{46080CA7-7CB8-3A55-A72E-8E50ECA4D4FC}\InProcServer32	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{46080CA7-7CB8-3A55-A72E-8E50ECA4D4FC}\InProcServer32\
6091	0001LXQEN	Classes\CLSID\{0C1A0EF4-B44F-4179-BDDD-31C27DE7A6D1}\InProcServer32	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{0C1A0EF4-B44F-4179-BDDD-31C27DE7A6D1}\InProcServer32\%
...	...	...	...
12168	0001LXQEN	Classes\CLSID\{B92E345D-F52D-41F3-B562-081BC772E3B9}\InProcServer32	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{B92E345D-F52D-41F3-B562-081BC772E3B9}\InProcServer32\%
12169	0001LXQEN	Classes\CLSID\{3050f6cd-98b5-11cf-bb82-00aa00bdce0b}\InProcServer32	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{3050f6cd-98b5-11cf-bb82-00aa00bdce0b}\InProcServer32\
12170	0001LXQEN	Classes\CLSID\{b27b520e-46db-4720-b9c5-5f80acab23a4}\InProcServer32	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{b27b520e-46db-4720-b9c5-5f80acab23a4}\InProcServer32\
12171	0001LXQEN	Classes\CLSID\{9456A480-E88B-43EA-9E73-0B2D9B71B1CA}\InProcServer32	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{9456A480-E88B-43EA-9E73-0B2D9B71B1CA}\InProcServer32\%
12172	0001LXQEN	Classes\CLSID\{32DA2B15-CFED-11D1-B747-00C04FC2B085}\InProcServer32	HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{32DA2B15-CFED-11D1-B747-00C04FC2B085}\InProcServer32\

6086 rows × 7 columns

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In [14]:

```
tasks[tasks['hostname'] == '0001LXQEN']
```

Out [14]:

	accountlogontype	accountrunlevel	certificateissuer	certificatesubject
126	TASK_LOGON_SERVICE_ACCOUNT	TASK_RUNLEVEL_HIGHEST	Microsoft Windows Production PCA 2011	Microsoft Windows
127	TASK_LOGON_SERVICE_ACCOUNT	TASK_RUNLEVEL_LUA	Microsoft Windows Production PCA 2011	Microsoft Windows \$(@%SystemRoot%\
128	TASK_LOGON_SERVICE_ACCOUNT	TASK_RUNLEVEL_LUA		This task e Ati
129	TASK_LOGON_SERVICE_ACCOUNT	TASK_RUNLEVEL_HIGHEST		Scans fault-tolerant v
130	TASK_LOGON_SERVICE_ACCOUNT	TASK_RUNLEVEL_HIGHEST	Microsoft Windows Production PCA 2011	Microsoft Windows Optimizes the storage tiers on all
...	...	...	...	...
248	TASK_LOGON_GROUP	TASK_RUNLEVEL_LUA		This task applies col
249	TASK_LOGON_GROUP	TASK_RUNLEVEL_HIGHEST		The Windows Dia Resolution ho resolutions for syste by the Diagnos triggered w Diagnosti appropriate user ses Policy Service is not
250	TASK_LOGON_SERVICE_ACCOUNT	TASK_RUNLEVEL_LUA	Microsoft Windows Production PCA 2011	Microsoft Windows This task trig followir
251	TASK_LOGON_SERVICE_ACCOUNT	TASK_RUNLEVEL_HIGHEST	Microsoft Windows Production PCA 2011	Microsoft Windows Launch I
252	TASK_LOGON_GROUP	TASK_RUNLEVEL_LUA	Microsoft Windows Production PCA 2011	Microsoft Windows Verifies AppUriHanc

127 rows x 14 columns

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```
In [17]: if 'hostname' in persistence_svc.columns and not persistence_svc.empty:
        print(persistence_svc[persistence_svc['hostname'] == '0001LXQEN'])
        else:
        print("persistence_svc table is empty - no service-based persistence data available in this dataset.")

persistence_svc table is empty - no service-based persistence data available in this dataset.

In [18]: print(persistence_reg[persistence_reg['hostname'] == '0001LXQEN'])
```

	hostname	\		keypath	\		path	\		text	type	\		username	valuenam
6087	0001LXQEN			Classes\CLSID\{4be5c3b3-53a2-4f2f-af9d-d26a5327eb92}\InProcServer32			HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{4be5c3b3-53a2-4f2f-af9d-d26a5327eb92}\InProcServer32\			C:\Windows\System32\IME\IMEKR\APPLETS\imkrskf.dll	REG_SZ		NT AUTHORITY\SYSTEM		
6088	0001LXQEN			Classes\CLSID\{ed9d80b9-d157-457b-9192-0e7280313bf0}\InProcServer32			HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{ed9d80b9-d157-457b-9192-0e7280313bf0}\InProcServer32\			%SystemRoot%\system32\zipfldr.dll	REG_EXPAND_SZ		NT AUTHORITY\SYSTEM		
6089	0001LXQEN		Wow6432Node\	Classes\CLSID\{3F037241-414E-11D1-A7CE-00A0C913F73C}\InProcServer32			HKEY_LOCAL_MACHINE\Software\Wow6432Node\Classes\CLSID\{3F037241-414E-11D1-A7CE-00A0C913F73C}\InProcServer32\			%SystemRoot%\System32\dmstyle.dll	REG_EXPAND_SZ		NT AUTHORITY\SYSTEM		
6090	0001LXQEN			Classes\CLSID\{46080CA7-7CB8-3A55-A72E-8E50ECA4D4FC}\InProcServer32			HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{46080CA7-7CB8-3A55-A72E-8E50ECA4D4FC}\InProcServer32\			C:\Windows\System32\mscoree.dll	REG_SZ		NT AUTHORITY\SYSTEM		
6091	0001LXQEN			Classes\CLSID\{0C1A0EF4-B44F-4179-BDDD-31C27DE7A6D1}\InProcServer32			HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{0C1A0EF4-B44F-4179-BDDD-31C27DE7A6D1}\InProcServer32\			%SystemRoot%\System32\InternetMailCsp.dll	REG_EXPAND_SZ		NT AUTHORITY\SYSTEM		
...	...			...			...			...	...		...		
12168	0001LXQEN			Classes\CLSID\{B92E345D-F52D-41F3-B562-081BC772E3B9}\InProcServer32			HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{B92E345D-F52D-41F3-B562-081BC772E3B9}\InProcServer32\			%SystemRoot%\system32\windowscodecs.dll	REG_EXPAND_SZ		NT AUTHORITY\SYSTEM		
12169	0001LXQEN			Classes\CLSID\{3050f6cd-98b5-11cf-bb82-00aa00bdce0b}\InProcServer32			HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{3050f6cd-98b5-11cf-bb82-00aa00bdce0b}\InProcServer32\			C:\Windows\System32\iepeers.dll	REG_SZ		NT AUTHORITY\SYSTEM		
12170	0001LXQEN			Classes\CLSID\{b27b520e-46db-4720-b9c5-5f80acab23a4}\InProcServer32			HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{b27b520e-46db-4720-b9c5-5f80acab23a4}\InProcServer32\			%SystemRoot%\system32\hgcp1.dll	REG_EXPAND_SZ		NT AUTHORITY\SYSTEM		
12171	0001LXQEN			Classes\CLSID\{9456A480-E88B-43EA-9E73-0B2D9B71B1CA}\InProcServer32			HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{9456A480-E88B-43EA-9E73-0B2D9B71B1CA}\InProcServer32\			%SystemRoot%\system32\windowscodecs.dll	REG_EXPAND_SZ		NT AUTHORITY\SYSTEM		
12172	0001LXQEN			Classes\CLSID\{32DA2B15-CFED-11D1-B747-00C04FC2B085}\InProcServer32			HKEY_LOCAL_MACHINE\Software\Classes\CLSID\{32DA2B15-CFED-11D1-B747-00C04FC2B085}\InProcServer32\			C:\Windows\System32\scrrun.dll	REG_SZ		NT AUTHORITY\SYSTEM		

	account	logontype	account	runlevel	\
6087	NT AUTHORITY\SYSTEM				
6088	NT AUTHORITY\SYSTEM				
6089	NT AUTHORITY\SYSTEM				
6090	NT AUTHORITY\SYSTEM				
6091	NT AUTHORITY\SYSTEM				
...	...				
12168	NT AUTHORITY\SYSTEM				
12169	NT AUTHORITY\SYSTEM				
12170	NT AUTHORITY\SYSTEM				
12171	NT AUTHORITY\SYSTEM				
12172	NT AUTHORITY\SYSTEM				

[6086 rows x 7 columns]

```
In [19]: print(tasks[tasks['hostname'] == '0001LXQEN'])
```

	account	logontype	account	runlevel	\
126	TASK_LOGON_SERVICE_ACCOUNT		TASK_RUNLEVEL_HIGHEST		
127	TASK_LOGON_SERVICE_ACCOUNT		TASK_RUNLEVEL_LUA		


```

128 TASK_LOGON_SERVICE_ACCOUNT TASK_RUNLEVEL_LUA
129 TASK_LOGON_SERVICE_ACCOUNT TASK_RUNLEVEL_HIGHEST
130 TASK_LOGON_SERVICE_ACCOUNT TASK_RUNLEVEL_HIGHEST
..
248 TASK_LOGON_GROUP TASK_RUNLEVEL_LUA
249 TASK_LOGON_GROUP TASK_RUNLEVEL_HIGHEST
250 TASK_LOGON_SERVICE_ACCOUNT TASK_RUNLEVEL_LUA
251 TASK_LOGON_SERVICE_ACCOUNT TASK_RUNLEVEL_HIGHEST
252 TASK_LOGON_GROUP TASK_RUNLEVEL_LUA

```

```

certificateissuer certificatesubject \
126 Microsoft Windows Production PCA 2011 Microsoft Windows
127 Microsoft Windows Production PCA 2011 Microsoft Windows
128
129
130 Microsoft Windows Production PCA 2011 Microsoft Windows
..
248
249
250 Microsoft Windows Production PCA 2011 Microsoft Windows
251 Microsoft Windows Production PCA 2011 Microsoft Windows
252 Microsoft Windows Production PCA 2011 Microsoft Windows

```

```

comment \
126
127
$(%SystemRoot%\system32\invagent.dll -702)
128
This task enrolls a certificate for Attestation Identity Key.
129
Scans fault-tolerant volumes for fast crash recovery
130
Optimizes the placement of data in storage tiers on all tiered storage spaces in the system.
..
...
248
This task applies color calibration settings.
249 The Windows Diagnostic Infrastructure Resolution host enables interactive resolutions for system problems d
ected by the Diagnostic Policy Service. It is triggered when necessary by the Diagnostic Policy Service in the
appropriate user session. If the Diagnostic Policy Service is not running the task will not run
250
This task triggers a system reboot following update installation.
251
Launch language cleanup tool
252
Verifies AppUriHandler host registrations.

```

```

creator \
126
127 $(%SystemRoot%\system32\invagent.dll -701)
128 Microsoft Corporation
129 Microsoft Corporation
130 Microsoft Corporation
..
248 Microsoft Corporation
249 Microsoft Corporation
250 Microsoft Corporation
251 Microsoft Corporation
252 Microsoft Corporation

```

```

description \
126 The file is signed and the signature was verified.
127 The file is signed and the signature was verified.
128
129
130 The file is signed and the signature was verified.
..
248
249
250 The file is signed and the signature was verified.
251 The file is signed and the signature was verified.
252 The file is signed and the signature was verified.

```

```

execarguments \
126 -z
127 -maintenance
128
129
130 -c -h -g -# -m 8 -i 13500
..
248
249

```

250
251
252

```
execprogrampath    hostname  \
126      %windir%\system32\disksnapshot.exe  0001LXQEN
127      %windir%\system32\compattelrunner.exe  0001LXQEN
128      0001LXQEN
129      0001LXQEN
130      %windir%\system32\defrag.exe  0001LXQEN
..      ...
248      0001LXQEN
249      0001LXQEN
250      %systemroot%\system32\MusNotification.exe  0001LXQEN
251      %windir%\system32\lpremove.exe  0001LXQEN
252      %windir%\system32\AppHostRegistrationVerifier.exe  0001LXQEN

name signatureexists signatureverified  \
126      Diagnostics      true      true
127      ProgramDataUpdater      true      true
128      AikCertEnrollTask
129      Data Integrity Scan for Crash Recovery
130      Storage Tiers Optimization      true      true
..      ...
248      Calibration Loader
249      ResolutionHost
250      Reboot      true      true
251      LPRemove      true      true
252      appuriverifierinstall      true      true

virtualpath
126      \Microsoft\Windows\DiskFootprint\Diagnostics
127      \Microsoft\Windows\Application Experience\ProgramDataUpdater
128      \Microsoft\Windows\CertificateServicesClient\AikCertEnrollTask
129      \Microsoft\Windows\Data Integrity Scan\Data Integrity Scan for Crash Recovery
130      \Microsoft\Windows\Storage Tiers Management\Storage Tiers Optimization
..      ...
248      \Microsoft\Windows\WindowsColorSystem\Calibration Loader
249      \Microsoft\Windows\WDI\ResolutionHost
250      \Microsoft\Windows\UpdateOrchestrator\Reboot
251      \Microsoft\Windows\MUI\LPRemove
252      \Microsoft\Windows\ApplicationData\appuriverifierinstall
```

[127 rows x 14 columns]

Findings Summary

Compromised machine identified: 0001LXQEN

Key evidence

1. SYSTEM-privilege execution of a user-located batch script

- Process: cmd.exe (PID 5612), running as NT AUTHORITY\SYSTEM
- Command: C:\Windows\System32\cmd.exe /C C:\Users\timotlopez\AppData\Local\Microsoft\WindowsApps\run_pd.bat
- This is highly anomalous: WindowsApps is a directory reserved for signed Microsoft Store (UWP) application stubs, not user-created batch scripts. A .bat file placed in this location and executed by a SYSTEM-level process is a strong indicator of **privilege escalation and/or a persistence mechanism**, since it grants attacker-controlled code the highest level of privilege on the machine.

2. High-privilege accounts logged on to the same host

- timotlopez (member of the domain admins group) is logged on to 0001LXQEN .
- Administrator (a domain admin account) is also logged on to 0001LXQEN .
- Concentration of privileged/domain-admin sessions on this specific host increases the potential impact of the compromise (domain-wide risk).

3. Host comparison

- Both hosts (0000DQQUEE and 0001LXQEN) share a very similar baseline of running processes and installed software (Avast, CyberGhost VPN, OneDrive, etc.).
- The SYSTEM-privileged cmd.exe invoking a script from a user profile path was **only observed on 0001LXQEN** , making this the differentiating factor that points to this host as compromised rather than the baseline/clean host.

Conclusion

Host **0001LXQEN** shows a credible indicator of compromise: a SYSTEM-level process executing a batch script from a non-standard, user-writable location, combined with active domain-admin-level sessions on the same machine. This is consistent with an attacker having achieved privilege escalation and/or established persistence on this host.

(See the accompanying report for the Executive Summary, IOC explanations in plain language, recommendations, limitations, and assumptions.)